DATA STRUCTURE AND PROGRAM DESIGN LAB – 03

3. Design, Develop and Implement a menu driven Program in C for the following operations on QUEUE of Characters (Array Implementation of Queue with maximum size MAX)
a. Insert an Element on to QUEUE
b. Delete an Element from QUEUE
c. Demonstrate Overflow and Underflow situations on QUEUE
d. Display the status of QUEUE
e. Exit
Support the program with appropriate functions for each of the above operations.

SAMPLE OUTPUT:

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 1
Enter character to insert: A

A inserted into queue.
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 1
Enter character to insert: B

B inserted into queue.
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 1
Enter character to insert: C

C inserted into queue.
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 3

Queue elements are:
A B C
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 2

A deleted from queue.
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 3

Queue elements are:
B C
```

```
--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 2

B deleted from queue.

--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit
```

```
Enter your choice: 2

C deleted from queue.

--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 2

Queue Underflow!
```

```
Queue Underflow!

--- QUEUE MENU ---
1. Insert
2. Delete
3. Display Queue
4. Exit

Enter your choice: 4
```